

ESM 120/MATH 128 Syllabus

Welcome to the Motivated Mathematical Mindset with professors Medina and Mendes (M^5)!

Instructors

- Elsa Medina emedina@calpoly.edu
- Tony Mendes aamendes@calpoly.edu

Course tutors

- Xavier Garcia
- Ava Silverstein
- Diego Ortiz

Course objectives

Many people view mathematics as memorizing formulas, following step-by-step procedures, and completing tedious exercises, but not us! Mathematics is creative, exploratory, and deeply connected to reasoning, communication, and discovery. We are excited to spend the summer with you, helping you develop a new perspective on what mathematics can be.

At the same time, we will cover many topics traditionally included in a Precalculus course. We want you to enter Calculus this Fall with stronger mathematical preparation than you have today. Our ultimate goal is to prepare you for success during your first year at Cal Poly and beyond.

This class can be a great deal of fun, especially when everyone is working together toward a common goal. Our expectations for you are:

1. You are engaged and excited to participate this summer.
2. You are using this course to fully prepare for the rigors of a university level mathematics course this Fall.
3. You recognize that Summer Institute, attending Cal Poly, and ultimately graduating from Cal Poly can be life-changing experiences, and you are committed to taking full advantage of these opportunities.
4. You approach being confused and tackling unfamiliar problems with resilience, confidence, and a willingness to make mistakes as part of the learning process.

Office hours

Professors Medina and Mendes will be at lunch with you most Mondays through Thursdays. We would greatly enjoy talking with you there about mathematics or just about anything else.

Mathematics topics

Our plans include these topics (with some special fun surprise topics added along the way):

- polynomial functions (roots, graphing, factoring)
- rational functions (roots, graphing)
- radical functions
- exponential functions and exponents

- functions (1-1, onto)
- Absolute values and solving inequalities
- Sigma notation and summation
- Circles and triangles and trigonometric functions

Grading

We will have almost daily quizzes and we will have a midterm on Thursday, July 23rd. These exams will be graded at the university level as if you were in one of our Calculus sections in the Fall, but the scores on your exams will not affect whether or not you earn credit in the Summer Institute course.

Credit in this course is determined by class participation. If you do what the instructors ask and are an active and enthusiastic participant, then you will earn credit. If you are in danger of not receiving credit because you are not treating this summer institute class with professionalism, then Professor Medina or Mendes will let you know early enough so that you have time to make the necessary adjustments.

Academic honesty

Please take pride in yourself and your work, practice integrity, and respect the efforts of your classmates and your professors. We're all better for it!

Diversity

We are committed to providing an atmosphere that is welcoming to everyone. It is our intent that students from diverse backgrounds and perspectives will be well served by this course.